



AMAL JYOTHI
COLLEGE OF ENGINEERING
A U T O N O M O U S
KANJIRAPPALLY

AUREON DATABASE TABLE DESIGN & NORMALIZATION

Scrum Master

Rony Tom

Assistant Professor

Department of Computer Applications

DEPARTMENT OF
COMPUTER APPLICATIONS

AUREON Database Table Design & BCNF Normalization

Role Table

Purpose: Stores the different user roles (Business Architect, Project Manager, Team Lead, Developer, Tester) used for role-based access control.

tbl_role

Field	Data Type	Key	Description
role_id (PK)	INT	Primary Key (PK)	Description of role_id (PK).
role_name	VARCHAR(50)	-	Description of role_name.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification:  Boyce-Codd Normal Form (BCNF)

- role_id → role_name, is_active, is_deleted

1NF: All attributes contain single atomic scalar values.

2NF: PK is single-column (role_id); no partial dependencies exist.

3NF: No non-key attribute determines any other non-key attribute (no transitive FDs).

BCNF: The sole determinant (role_id) is a superkey/candidate key.

User Table

Purpose: Stores user registration and login information along with profile details, role, and team assignment.

tbl_user

Field	Data Type	Key	Description
user_id (PK)	INT	Primary Key (PK)	Description of user_id (PK).
user_name	VARCHAR(100)	-	Description of user_name.
user_email	VARCHAR(100)	-	Description of user_email.
user_password	VARCHAR(255)	-	Description of user_password.
user_phone	VARCHAR(15)	-	Description of user_phone.
user_employee_id	VARCHAR(30)	-	Description of user_employee_id.
role_id (FK)	INT	Foreign Key (FK)	Description of role_id (FK).
team_id (FK)	INT	Foreign Key (FK)	Description of team_id (FK).
user_status	ENUM	-	Description of user_status.
user_created_at	DATETIME	-	Description of user_created_at.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: ✓ Boyce-Codd Normal Form (BCNF)

- $user_id \rightarrow user_name, user_email, user_password, user_phone, user_employee_id, role_id, team_id, user_status, user_created_at, is_active, is_deleted$
- $user_email \rightarrow user_id$ (Candidate Key)
- $user_employee_id \rightarrow user_id$ (Candidate Key)

1NF: All columns contain atomic attributes.

2NF: Primary key $user_id$ is non-composite; no partial key dependencies.

3NF: Non-prime attributes depend solely on candidate keys.

BCNF: All functional determinants ($user_id, user_email, user_employee_id$) are Candidate Keys.

Project Table

Purpose: Stores software project information including project details, repository, project manager, and business architect.

tbl_project

Field	Data Type	Key	Description
project_id (PK)	INT	Primary Key (PK)	Description of project_id (PK).
project_name	VARCHAR(100)	-	Description of project_name.
project_description	TEXT	-	Description of project_description.
project_start_date	DATE	-	Description of project_start_date.
project_end_date	DATE	-	Description of project_end_date.
project_repository_url	VARCHAR(255)	-	Description of project_repository_url.
project_status	VARCHAR(30)	-	Description of project_status.
business_architect_id (FK)	INT	Foreign Key (FK)	Description of business_architect_id (FK).
project_manager_id (FK)	INT	Foreign Key (FK)	Description of project_manager_id (FK).
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: ✓ Boyce-Codd Normal Form (BCNF)

- $project_id \rightarrow project_name, project_description, project_start_date, project_end_date, project_repository_url, project_status, business_architect_id, project_manager_id, is_active, is_deleted$

1NF: All values are scalar and atomic.

2NF: Simple primary key $project_id$.

3NF: Direct non-transitive dependency on primary key.

BCNF: The functional determinant $project_id$ is a superkey.

Team Table

Purpose: Stores team details and associates each team with a project and team lead.

tbl_team

Field	Data Type	Key	Description
team_id (PK)	INT	Primary Key (PK)	Description of team_id (PK).

Field	Data Type	Key	Description
project_id (FK)	INT	Foreign Key (FK)	Description of project_id (FK).
team_name	VARCHAR(100)	-	Description of team_name.
team_type	VARCHAR(50)	-	Description of team_type.
team_lead_id (FK)	INT	Foreign Key (FK)	Description of team_lead_id (FK).
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: Boyce-Codd Normal Form (BCNF)

- team_id* → *project_id*, *team_name*, *team_type*, *team_lead_id*, *is_active*, *is_deleted*

1NF: Atomic values.

2NF: Single column key *team_id*.

3NF: No transitive FDs.

BCNF: *team_id* is a Candidate Key.

Sprint Table

Purpose: Stores sprint information such as sprint goals, schedule, priority, and sprint status.

tbl_sprint

Field	Data Type	Key	Description
sprint_id (PK)	INT	Primary Key (PK)	Description of sprint_id (PK).
project_id (FK)	INT	Foreign Key (FK)	Description of project_id (FK).
sprint_name	VARCHAR(50)	-	Description of sprint_name.
sprint_goal	TEXT	-	Description of sprint_goal.
sprint_start	DATE	-	Description of sprint_start.
sprint_end	DATE	-	Description of sprint_end.
sprint_priority	ENUM	-	Description of sprint_priority.
sprint_status	VARCHAR(30)	-	Description of sprint_status.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: Boyce-Codd Normal Form (BCNF)

- sprint_id* → *project_id*, *sprint_name*, *sprint_goal*, *sprint_start*, *sprint_end*, *sprint_priority*, *sprint_status*, *is_active*, *is_deleted*

1NF: Atomic domains.

2NF: No partial key dependency.

3NF: Non-key attributes depend only on *sprint_id*.

BCNF: *sprint_id* is a Candidate Key.

Task Table

Purpose: Stores all tasks assigned to developers, including deadlines, progress, and task status.

tbl_task

Field	Data Type	Key	Description
task_id (PK)	INT	Primary Key (PK)	Description of task_id (PK).
sprint_id (FK)	INT	Foreign Key (FK)	Description of sprint_id (FK).
assigned_by	INT	-	Description of assigned_by.
assigned_to	INT	-	Description of assigned_to.
team_id	INT	-	Description of team_id.
task_title	VARCHAR(150)	-	Description of task_title.
task_description	TEXT	-	Description of task_description.
task_priority	ENUM	-	Description of task_priority.
task_deadline	DATE	-	Description of task_deadline.
task_progress	INT	-	Description of task_progress.
task_status	VARCHAR(30)	-	Description of task_status.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: Boyce-Codd Normal Form (BCNF)

- task_id* → *sprint_id*, *assigned_by*, *assigned_to*, *team_id*, *task_title*, *task_description*, *task_priority*, *task_deadline*, *task_progress*, *task_status*, *is_active*, *is_deleted*

1NF: Complies with atomicity.

2NF: Single-attribute primary key *task_id*.

3NF: All attributes depend directly on *task_id*.

BCNF: Determinant *task_id* is a superkey/candidate key.

Repository Table

Purpose: Stores Git repository information for each software project used for code analysis.

tbl_repository

Field	Data Type	Key	Description
repository_id (PK)	INT	Primary Key (PK)	Description of repository_id (PK).
project_id (FK)	INT	Foreign Key (FK)	Description of project_id (FK).
repository_name	VARCHAR(100)	-	Description of repository_name.
repository_url	VARCHAR(255)	-	Description of repository_url.
repository_branch	VARCHAR(100)	-	Description of repository_branch.
repository_clone_status	VARCHAR(30)	-	Description of repository_clone_status.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: **Boyce-Codd Normal Form (BCNF)**

- *repository_id* → *project_id*, *repository_name*, *repository_url*, *repository_branch*, *repository_clone_status*, *is_active*, *is_deleted*

1NF: Atomic attributes.

2NF: Single column PK *repository_id*.

3NF: Direct functional dependency.

BCNF: *repository_id* is a Candidate Key.

Code Analysis Table

Purpose: Stores the results of static code analysis such as maintainability, complexity, technical debt, documentation quality, and overall code quality score.

tbl_code_analysis

Field	Data Type	Key	Description
analysis_id (PK)	INT	Primary Key (PK)	Description of analysis_id (PK).
repository_id (FK)	INT	Foreign Key (FK)	Description of repository_id (FK).
analysis_date	DATETIME	-	Description of analysis_date.
analysis_maintainability	FLOAT	-	Description of analysis_maintainability.
analysis_complexity	FLOAT	-	Description of analysis_complexity.
analysis_duplication	FLOAT	-	Description of analysis_duplication.
analysis_technical_debt	FLOAT	-	Description of analysis_technical_debt.
analysis_documentation	FLOAT	-	Description of analysis_documentation.
analysis_overall_score	FLOAT	-	Description of analysis_overall_score.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: **Boyce-Codd Normal Form (BCNF)**

- *analysis_id* → *repository_id*, *analysis_date*, *analysis_maintainability*, *analysis_complexity*, *analysis_duplication*, *analysis_technical_debt*, *analysis_documentation*, *analysis_overall_score*, *is_active*, *is_deleted*

1NF: Atomic metrics.

2NF: Primary key is single-valued *analysis_id*.

3NF: Metrics depend strictly on *analysis_id*.

BCNF: *analysis_id* is a superkey / Candidate Key.

Dashboard Table

Purpose: Stores project dashboard metrics including health score, sprint completion, task completion, and repository status.

tbl_dashboard

Field	Data Type	Key	Description
dashboard_id (PK)	INT	Primary Key (PK)	Description of dashboard_id (PK).
project_id (FK)	INT	Foreign Key (FK)	Description of project_id (FK).

Field	Data Type	Key	Description
dashboard_health_score	FLOAT	-	Description of dashboard_health_score.
dashboard_sprint_completion	FLOAT	-	Description of dashboard_sprint_completion.
dashboard_task_completion	FLOAT	-	Description of dashboard_task_completion.
dashboard_repository_status	VARCHAR(30)	-	Description of dashboard_repository_status.
dashboard_updated_at	DATETIME	-	Description of dashboard_updated_at.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: **Boyce-Codd Normal Form (BCNF)**

- dashboard_id* → *project_id*, *dashboard_health_score*, *dashboard_sprint_completion*, *dashboard_task_completion*, *dashboard_repository_status*, *dashboard_updated_at*, *is_active*, *is_deleted*

1NF: Atomic values.

2NF: Single column PK *dashboard_id*.

3NF: No transitive dependency.

BCNF: *dashboard_id* is a Candidate Key.

Report Table

Purpose: Stores notifications sent to users regarding task assignments, project updates, reminders, and alerts.

tbl_report

Field	Data Type	Key	Description
report_id (PK)	INT	Primary Key (PK)	Description of report_id (PK).
project_id (FK)	INT	Foreign Key (FK)	Description of project_id (FK).
report_type	VARCHAR(50)	-	Description of report_type.
report_generated_by	INT	-	Description of report_generated_by.
report_generated_date	DATETIME	-	Description of report_generated_date.
report_file_path	VARCHAR(255)	-	Description of report_file_path.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

Normalization Verification: **Boyce-Codd Normal Form (BCNF)**

- report_id* → *project_id*, *report_type*, *report_generated_by*, *report_generated_date*, *report_file_path*, *is_active*, *is_deleted*

1NF: Atomic attributes.

2NF: Single column PK *report_id*.

3NF: Non-key attributes depend only on *report_id*.

BCNF: *report_id* is a Candidate Key.

Notification Table

Purpose: Stores notifications sent to users regarding task assignments, project updates, reminders, and alerts.

tbl_notification

Field	Data Type	Key	Description
notification_id (PK)	INT	Primary Key (PK)	Description of notification_id (PK).
user_id (FK)	INT	Foreign Key (FK)	Description of user_id (FK).
notification_title	VARCHAR(150)	-	Description of notification_title.
notification_message	TEXT	-	Description of notification_message.
notification_status	ENUM	-	Description of notification_status.
notification_created_at	DATETIME	-	Description of notification_created_at.
is_active	BOOLEAN	-	Indicates whether the record is active (1 = Active, 0 = Inactive).
is_deleted	BOOLEAN	-	Indicates whether the record has been soft deleted (0 = Not Deleted, 1 = Deleted).

- Normalization Verification:**  **Boyce-Codd Normal Form (BCNF)**
- notification_id* → *user_id*, *notification_title*, *notification_message*, *notification_status*, *notification_created_at*, *is_active*, *is_deleted*
- 1NF:** Atomic fields.
- 2NF:** Single column PK *notification_id*.
- 3NF:** Direct non-transitive dependency.
- BCNF:** *notification_id* is a Candidate Key.